

# Agora Garden: An AI-Powered Personal Learning Planner

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# Agenda

- 1 Project Background and Objective
- 2 Our Solutions
- 3 Live Demo

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## Project Background and Objective

# Inconsistent skill requirements across similar functions and roles make it difficult for students and career-switching professionals to build a clear and effective learning path

- Experience with or knowledge of operational data platforms and observability tools, such as AIOps, Splunk, telemetry systems, DevOps tools, and Dynatrace
- Experience applying data analytics techniques, including anomaly detection, forecasting, and data engineering
- Proficiency in data visualization and dashboard development using tools such as Power BI, R, Shiny, or Splunk
- Ability to analyze complex datasets and translate insights into actionable business recommendations
- Hands-on experience with machine learning projects, including model development, deployment, or operationalization is a plus

## Preferred Experience with Analytical Tools/Applications:

Unix/Linux

Big Data Ecosystem: Hadoop, Spark, MapReduce, SQL, Hiv

Scientific Computing: R, Python, C++, Java, Scala,

High-Performance Parallel and Distributing Computing

Experience working with statistical packages (e.g., R, SAS, Stata, MATLAB, etc.).

Minimum Qualifications: - Currently pursuing an Undergraduate/Master in Computer Science, Computer Engineering or a related technical discipline - Experience in data science roles, preferably working on mobile application performance analysis and user experience optimization - Experience working with creation-oriented user behavior - Proficient in SQL and Python; experience with big data tools such as Hive or Spark. Preferred Qualifications: - Experience within

Programming Languages: Proficiency in SQL, Python, R, and cloud-based platforms like Databricks.

Analytical Skills: Expertise in data mining, statistical modeling, and machine learning.

Reporting and Visualization Tools: Experience with Excel, PowerPoint, Tableau, Power BI, and similar

## For students, traditional career centers struggle with capacity, accessibility, and specificity constraints

1

### Capacity: Career Center Staffing Falls Short of Student Demand

- The median U.S. college career center operates with only **7.0 full-time-equivalent staff**
- Personnel costs account for around **87% of the total budget** → hard to increase personnel

2

### Accessibility: Support Is Constrained by Office Hours and Advisor Availability

- Limited to fixed operating hours; unavailable during **evenings, weekends, and holidays**
- Depends on advisor schedule and finite appointment capacity, limiting **simultaneous access**

3

### Specificity: Generalized Guidance Falls Short of Role-Specific Career Needs

- Services primarily focus on **broad career skills** such as resume writing and interviewing
- Advisors may lack deep, up-to-date insights into **specific industries, functions, and roles**

What students need instead:

Scalable

On-Demand

Role-Specific

## Project Background and Objective

For working professionals, companies increasingly invest in employee learning and skill development to build workforce capabilities

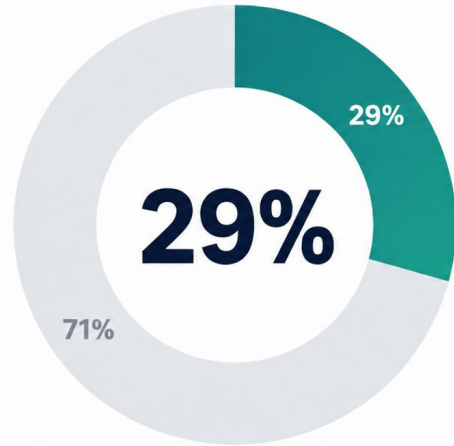


udemy business™

coursera for business

## However, employees are often given learning resources without a structured roadmap to guide skill progression

Employees struggle to turn available resources into skills



Only **29%** of employees say their L&D function helps them learn effectively from available people, experiences or content.

HR struggles to provide clear career pathways

Nearly **90%**

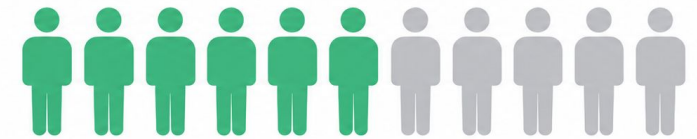
of HR leaders say career paths are unclear for many employees.



~ 9 out of 10 HR leaders

Two-thirds (**67%**)

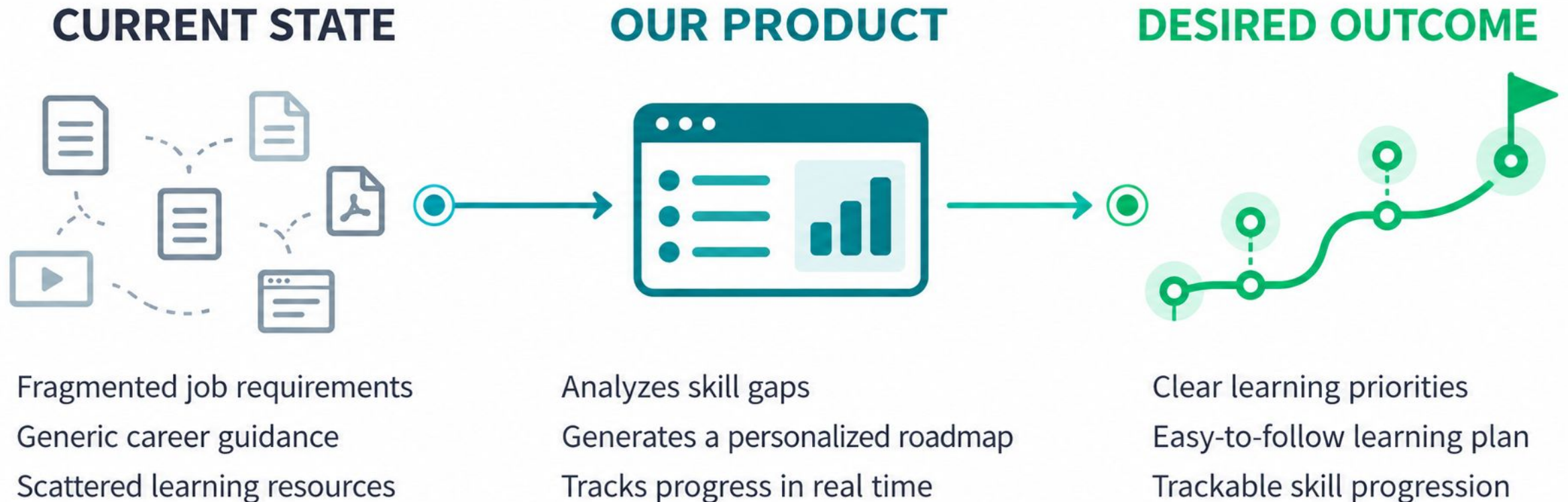
say career paths are not compelling for many employees.



~ 2 out of 3 HR leaders

Employers need a structured way to translate learning resources into **clear, role-specific skill development pathways**

**We help students and working professionals build targeted learning paths – whether they are preparing for their careers or advancing in their current role**



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Our product is a web-based platform that generates personalized learning plans and enables users to track their progress in real time

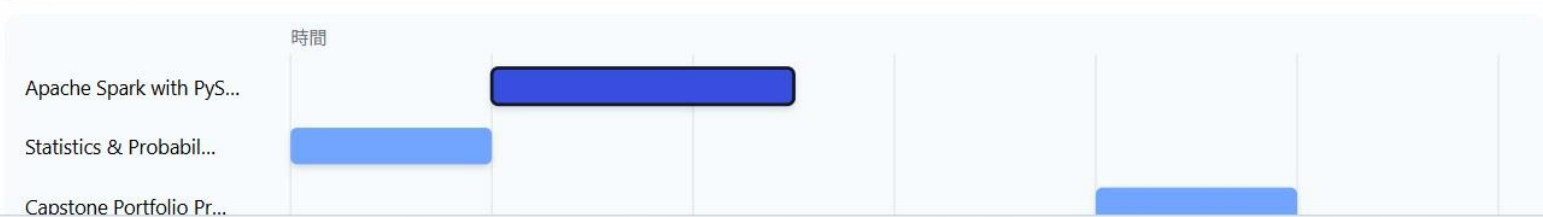
How can I help you, CHIH-FAN LIN?

Summary

Chih-Fan has strong Python, SQL, and ML project experience but lacks Spark, Airflow, Kafka, A/B testing, and formal statistics—skills appearing in most data scientist job descriptions. Priority: build Spark and statistics/A-B testing foundations first, then orchestration/streaming, then a capstone project and interview prep.

Learning Roadmap

STAGE  
全部



Learning Details

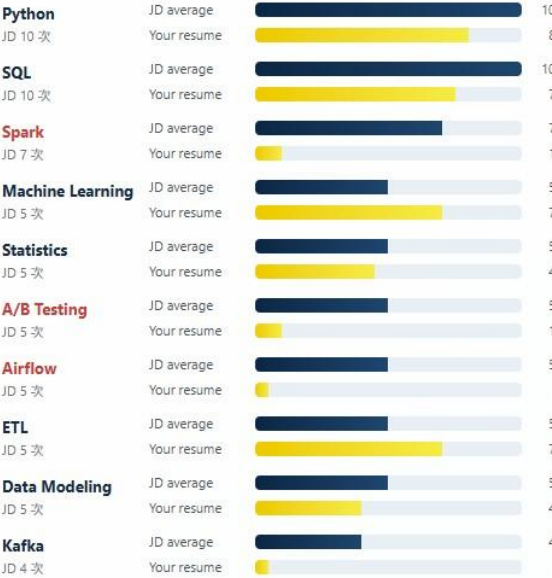
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Your Career Goal

Resume (PDF / DOCX / TXT)  
選擇檔案 Resume(Data Engineer).pdf

Top JD Skills vs You

依 10 則職缺 JD 統計：JD average 為該技能出現在職缺中的比例；Your resume 為履歷技能評分 (0-100)。



# Our product is powered by IFM’s 375B-A23B model for AI-driven skill analysis and learning roadmap generation

Model in K2 Horizon

375B-A23B

Prompt

We are currently participating in a hackathon and plan to use the K2 Horizon API.

**Problem We Want to Solve**

Our target users are students who have just entered university and already have a specific career goal in mind—for example, they may want to work in a particular industry or eventually apply for a specific job at a specific company. However, they often don't know how to strategically plan their university years to build the skills and experience needed to reach that goal.

Our idea is to build a web application that allows users to upload their resume and background information. The system will then analyze their current experience and, based on their target career goal, automatically generate a personalized roadmap for their university years.

The roadmap will include three main areas:

- 1. University coursework — What courses the student should consider taking to develop the required knowledge and skills.
- 2. External resources — Relevant resources outside the university, such as Coursera courses, certifications, workshops, competitions, or other learning opportunities.
- 3. Side projects — Projects the student could build to gain practical experience and strengthen their resume.

The system will also perform web searches to identify the skills currently required for the target position and use those requirements to build the roadmap.

**Target Users and Business Model**

Our original target users were students, with the idea of partnering with universities and providing this service as a career-development resource for their students. However, after discussing the idea with our team, we realized that there may also be a B2B opportunity with companies.

Many companies are currently encouraging employees to take ownership of their professional development and continuously learn new skills. Our platform could potentially be used by companies as an internal career-development or upskilling tool. Employees could specify the position or career path they are interested in, and the system could show them what skills they need to develop and how they can improve themselves over time.

Therefore, we are currently considering two potential customer segments:

- Universities — Helping students plan their academic and career development.
- Companies — Helping employees identify skill gaps and create personalized upskilling/career-development plans.

**Refined User Flow**

After further discussion, we refined the workflow.

When a user uploads their resume, they must also specify the job position they are interested in. The industry and company are optional.

For example:

- Required: "Software Engineer"
- Optional: "AI industry"
- Optional: "Google"

The system will then use the user's target position as the primary reference point.

It will:

1. Analyze the user's resume and background.
2. Search the web for current job descriptions (JDs) related to the user's target position.
3. Identify the most important skills currently required for that position.
4. Compare those requirements with the user's existing skills and experience.
5. Identify the user's skill gaps.
6. Generate a personalized roadmap to help the user eventually become qualified for the target position.

**Backend Output**

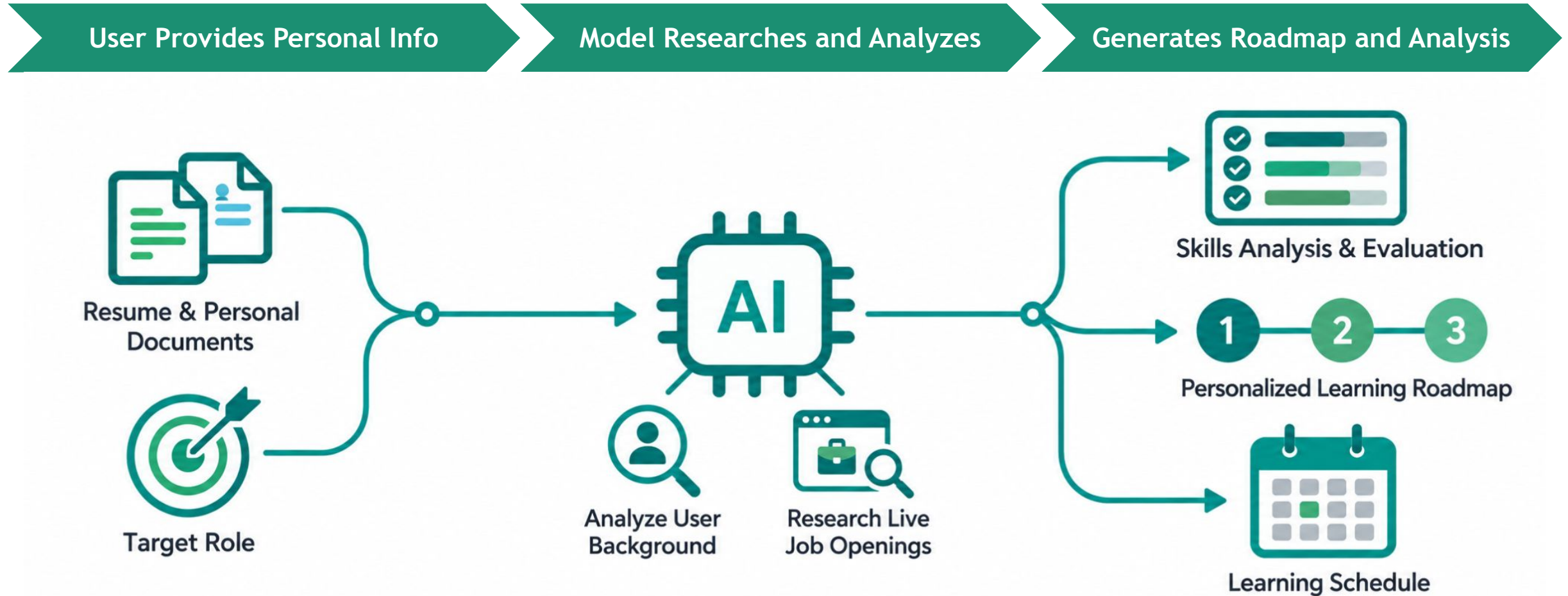
I am currently responsible for the backend, while my teammate is responsible for the frontend and web design.

Therefore, I expect the backend to produce structured outputs that the frontend can directly consume.

The backend should generate a summary explaining the overall roadmap, including why certain courses, external resources, and side projects are recommended.

## Our Solutions

Our product analyzes real-time job openings against user's background and desired roles, identifies and evaluates skill gaps, and generates personalized learning roadmap



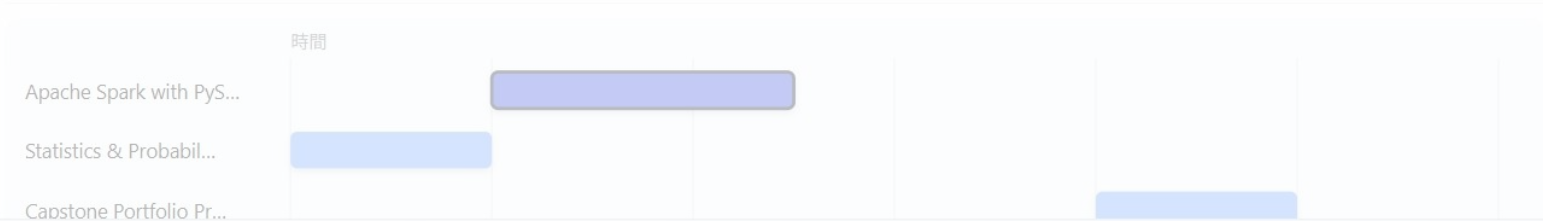
Users upload their resumes and other relevant documents, and select a target role, to receive a personalized learning roadmap and schedule

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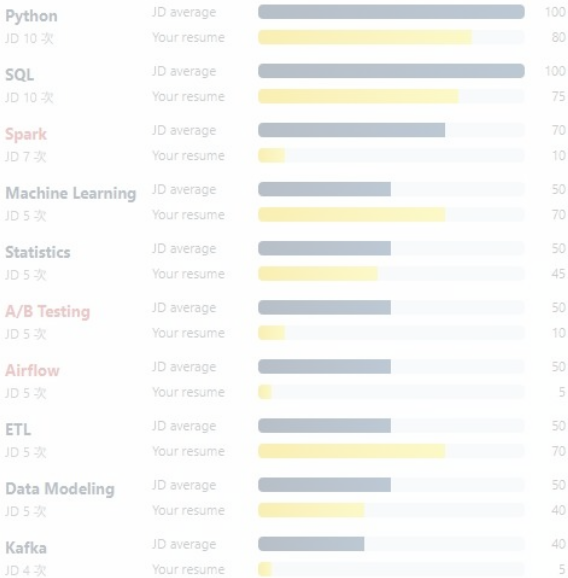
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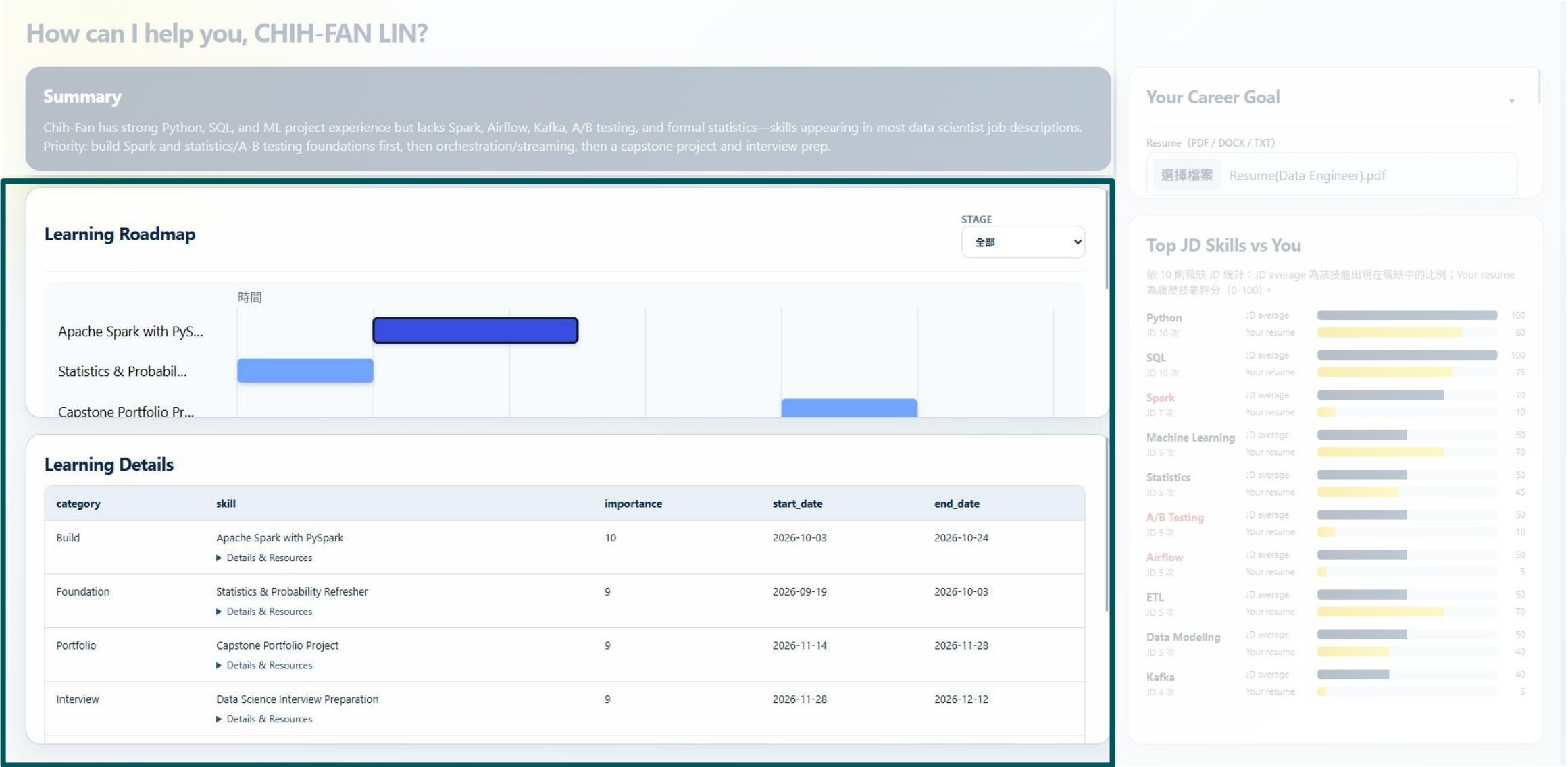
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# Timeline and table views simplifies tracking progress and managing learning schedules for users



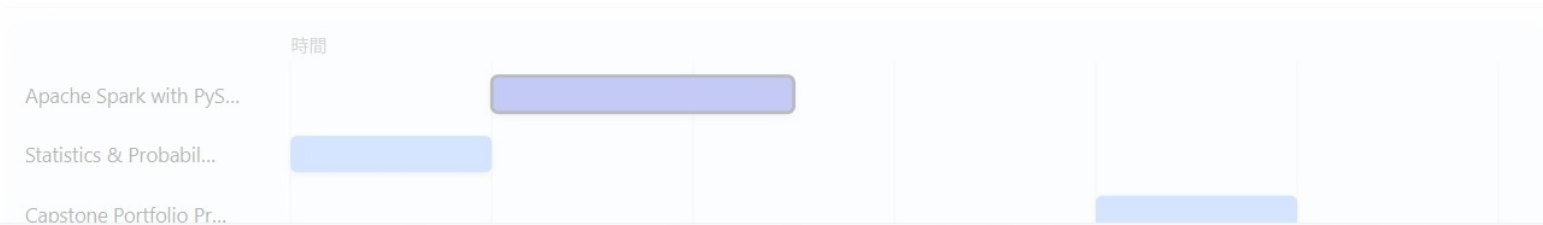
The skills section shows the most in-demand skills for users' target roles, and evaluates their proficiency in each skill based on their uploaded documents

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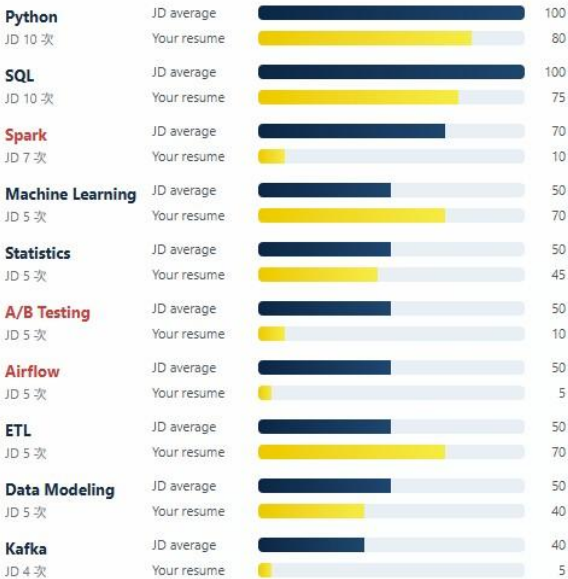
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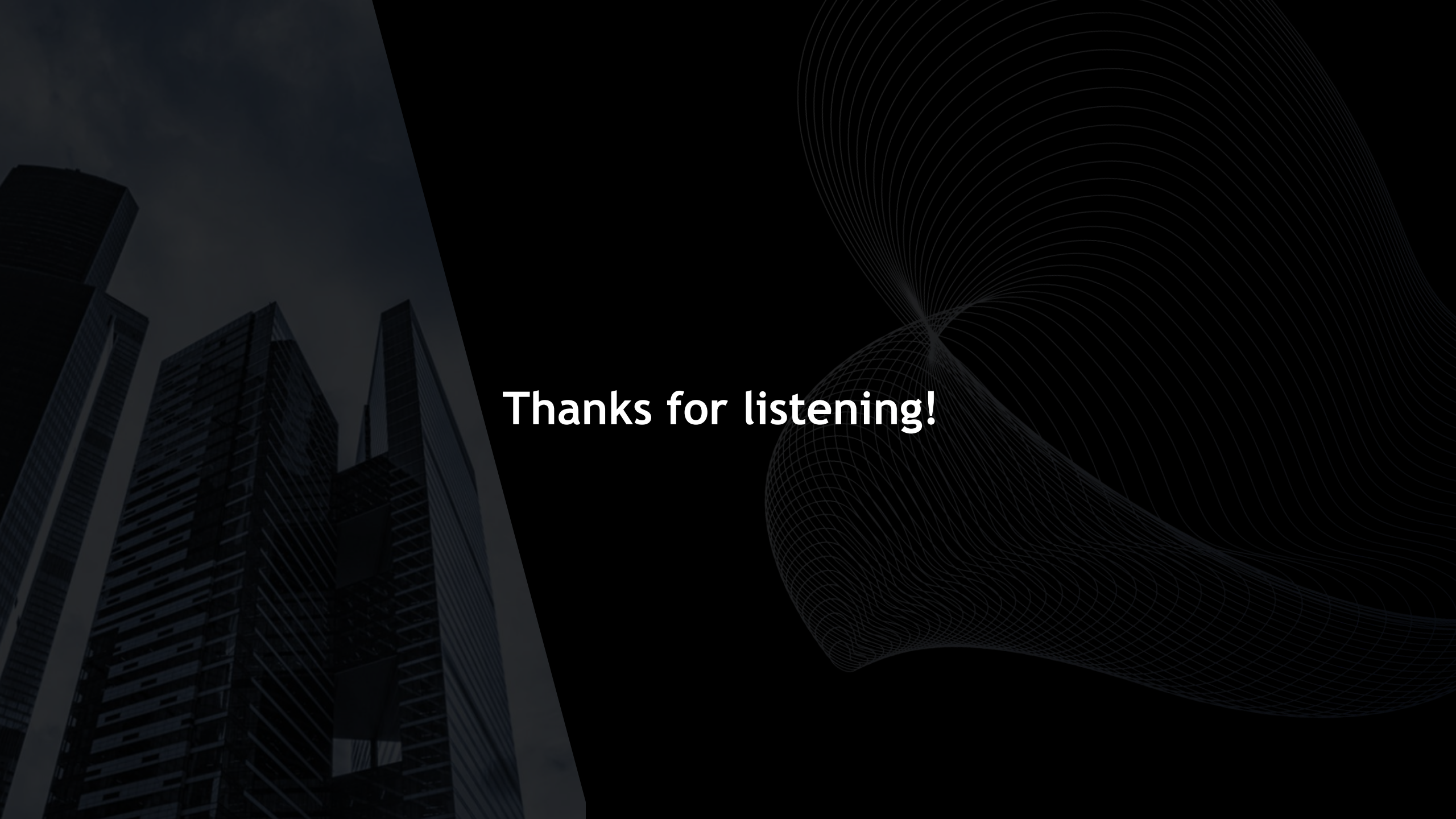
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The background is a dark, monochromatic composition. On the left, several tall, modern skyscrapers are visible, their forms partially obscured by the dark overlay. On the right, a large, intricate wireframe sphere is depicted, composed of numerous thin, curved lines that create a sense of depth and movement. The overall aesthetic is futuristic and architectural.

**Thanks for listening!**